Deploy Your Agent Harness to the Cloud

Mode 2 Phase 3 (Scale the Workforce), Chapter 7/9 — Revision & Viva-Prep Cheatsheet (English + Roman Urdu)
Source: agentfactory.panaversity.org/docs/deploying-agents-crash-course

EN: Core idea: the HARNESS is the control plane you own; the SANDBOX is the throwaway execution plane. Brain on Azure Container Apps, memory in Neon Postgres, files in Cloudflare R2, risky code in E2B/Cloudflare Sandbox.
Roman Urdu: Core idea: HARNESS control plane hai jo aap own karte; SANDBOX throwaway execution plane hai. Brain Azure Container Apps par, memory Neon Postgres mein, files Cloudflare R2 mein, risky code E2B/Cloudflare Sandbox mein.

1. The Harness/Sandbox Split

- Harness = brain (requests, agent loop, durable state, secrets). Sandbox = hands (workspace, execution, NO secrets access except Manifest-mounted).
- **2 anti-patterns (the 'original sin'):** running the harness inside the sandbox, or running agent-generated code inside the harness.
- 5 production properties a laptop lacks: multi-user, restart-surviving state, isolated code execution, secrets out of reach, observable/auditable runs.

2. The Five-Component Stack

- FastAPI (async-native, auto OpenAPI). Azure Container Apps (public address, autoscale-to-zero, secrets, blue/green revisions, observability).
- Neon Postgres: footgun -- the pooled endpoint silently drops search_path, so always schema-qualify (public.runs). Cloudflare R2: egress-free (reading your own files is free); presigned URLs give scoped access without root credentials.

3. The Execution Plane + Observability

- Sandbox needs 3 production-grade properties: isolation, ephemerality (task-fresh), fast provisioning. A Manifest controls what's mounted/on.
- **Capabilities.default() REPLACES the passed list** (footgun) -- concatenate to add, don't just pass new ones.
- Most production AI failures are OBSERVABILITY failures -- 4 surfaces (Application Insights, OTel traces, SDK traces, Phoenix) interconnect via a shared run_id.

4. Cost + What Deployment Doesn't Solve

- Model API cost is 90-98% of the bill at any scale; infrastructure is under 5%. Optimize the model, not the infra.
- **5 things you should never do:** exec(model_output) in the harness, root creds in the Manifest, skip scale-to-zero in dev, deploy without an eval suite, no rate limiting.
- Deployment does NOT solve: compliance certification, incident-response program, legal liability, prompt-injection behavior, or cost runaway on its own.

Key Terms Cheatsheet

One line each — English + Roman Urdu.

Term	English	Roman Urdu
Harness vs Sandbox	Control plane you own (secrets, state, loop) vs throwaway execution plane (no secrets).	Control plane jo aap own karte (secrets, state, loop) vs throwaway execution plane (secrets nahi).
Manifest	Workspace description -- what to mount, what to turn on.	Workspace description -- kya mount karna, kya on karna.
search_path footgun	Neon's pooled endpoint silently drops it -- always schema-qualify queries.	Neon ka pooled endpoint isay silently drop karta -- hamesha schema-qualify karo.
Presigned URL	Scoped, time-limited access to R2 without root credentials.	R2 tak scoped, time-limited access, root credentials ke bagair.
Shared run_id	Links logs across all 4 observability surfaces for one request.	Ek request ke 4 observability surfaces ko link karta.

Quick Self-Check

Question	Answer
Why must agent-generated code never run inside the harness?	That's the 'original sin' -- it would give untrusted generated code access to the harness's secrets and durable state.
What's the Neon pooled-endpoint footgun, and the fix?	It silently drops search_path -- always schema-qualify table names (e.g. public.runs) to avoid queries silently hitting the wrong schema.
Why is Capabilities.default() a footgun?	Passing a new capabilities list REPLACES the defaults instead of adding to them -- you must concatenate to keep default capabilities.
What share of the total bill is typically model API cost vs infrastructure?	Model API is 90-98% at any scale; infrastructure is under 5% -- cost optimization should target the model, not the infra.

Where This Leads / Sources

Leads to: Choosing Agentic Architectures, Payment-Enabled Agents. Full 9-decision deployment lab (D0-D9) in Part 5.

Source: docs/deploying-agents-crash-course/ (this repo), content originally via Zia Tutor AI connector.